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Effective Dielectric Function of a Metal-Dielectric Composite with Nonrandomly Distributed Particles (*).

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Summary. — The optical properties of a metal-dielectric binary composite with nonrandomly distributed particles were studied within the framework of the effective medium theory. All the particles, as usual, were assumed to be spherical and the nonrandomness of their distribution was achieved by forcing onto the system the correlation that prevents the metal particles from touching each other or the one that forces them to aggregate in pairs, either oriented at random or all alike. The drawbacks of the effective medium theory due to the assumption that the particles be very small were overcome by describing the scattering properties of the particles through their exact Mie amplitudes. Both kinds of correlation we considered turned out to have substantial effects on the onset of the percolation phenomena. The transition induced by the correlation of exclusion from a behaviour typical of the Bruggeman model to one more appropriate to the Maxwell-Garnett model was followed as a function of the exclusion radius. In the case of aggregation the onset of the percolation phenomena was found to be strongly influenced by the further introduction of an orientational order of the pairs.

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1. – Introduction.

The propagation of electromagnetic waves through a composite medium can be dealt with by the *effective medium theory* [1-3]. According to this approach one could

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replace the actual nonhomogeneous composite by an effective homogeneous medium without affecting the propagation provided that ε_{eff}, the dielectric function of the effective medium, is properly chosen. Indeed, ε_{eff} must be such that, when the composite is embedded into the effective medium, no scattering is experienced. Since ε_{eff} can be understood as an average of the local dielectric function of the composite, this latter should be not too coarsely granulated, *i.e.* its component particles should be smaller, and possibly much smaller, than the wavelength of the propagation radiation. Of course, the topology of the composite is of fundamental importance in determining ε_{eff}. In fact, if the medium is composed of very small particles that fill the whole space (the *aggregate structure model*) one gets the Bruggeman mixing rule, while, if the (still very small) particles of the composite are well separated and dispersed within a homogeneous matrix (the *isolated grain model*), one gets the Maxwell Garnett mixing rule [1]. Both these rules were obtained approximating the particles by spheres and thus neglecting their real shape.

As far as we know, no attempt has been made to modify the Bruggeman rule to include the possible effects of interparticle correlation or of the aggregation of the particles themselves. This is not surprising, in a sense, because the Bruggeman rule was originally devised in the static limit and the main concern of the researchers has been to extend it to finite frequencies and to test its range of applicability. Webman, Jortner and Cohen proposed a self-consistent extension of the Bruggeman rule to finite frequencies [4], and, by comparing the results of a numerical simulation with the predictions of this extended rule, found that a necessary condition for the validity of the effective medium theory is that the local fluctuations of the dielectric function lie in a well-defined range [3]. More precisely the condition found by these authors for a metal-dielectric binary composite reads $0.05 < |\varepsilon_{\text{M}}/\varepsilon_{\text{D}}| \leqslant 20$, where ε_{M} and ε_{D} are the dielectric functions of the metal and of the dielectric, respectively. The problem of the range of validity of the effective medium approximation has been resumed by Stroud and Pan [2] who introduced a rigorous method for extending the Bruggeman mixing rule to finite frequencies and also proved that substituting the actual medium with an effective one amounts to account, in an average sense, for the multiple scattering processes that occur within the composite. Stroud and Pan were also able to prove that their extension of the Bruggeman mixing rule can be applied to a composite of metal and of dielectric spherical particles provided that $x\,\text{Im}[\varepsilon_{\text{eff}}] \ll 1$, where $x = \omega\rho/c$ is the *size parameter* of any of the particles with radius ρ.

For most metal-dielectric composites both the conditions stated above can be hardly satisfied but at frequencies for which the metal is by itself not too strongly absorbing. As a general rule, however, even when these conditions are not satisfied, the effective medium theory still works satisfactorily provided that the volume fraction occupied by the metallic component is small; nevertheless this theory is unable to define correctly V^*, the volume fraction of the metallic component at which the percolation phenomena set on, and ε_{eff} thus assumes values typical of a metal. As an extreme case to be discussed later, the *dynamic effective medium approximation* (DEMA) proposed by Stroud and Pan [2] yields no acceptable result when $x = 0.1$ and $\omega = 0.01\omega_{\text{p}}$, where ω_{p} is the plasma frequency of the metal spheres.

The aim of this paper is to introduce the effect of the interparticle correlation and of the anisotropy of the particles themselves within the framework of the aggregate structure model as applied to a binary metal-dielectric composite at concentrations of the metallic component near to V^*. Our investigation is inserted on the effective medium theory in spite of the well-known facts that, in most cases, V^* is incorrectly

evaluated, and that, even when all the conditions of validity referred to above are satisfied, the theory is affected by the well-known unphysical behaviour of ε_{eff} as a function of V when $V^* \leqslant V \leqslant 1 - V^*$. Nevertheless, for our purposes, these drawbacks of the theory are of little importance, provided the effects we study are understood in a relative sense. It is more important, instead, to remove preliminarily the unwanted effects of any unnecessary approximation on which the Bruggeman mixing rule and its extensions are based. Therefore, in this paper we study the properties of the same metal-dielectric composite investigated by Stroud and Pan [2] and by Chýlek and Srivastava [5] but, unlike these authors, we describe the scattering properties of the component spherical particles by their exact Mie amplitudes (we will refer to this approach as Bruggeman-Mie model) including as many amplitudes are necessary to get fully convergent results. We anticipate here, as an example, that the Bruggeman-Mie model, when $x = 0.1$ and $\omega = 0.01$, yields results that are a smooth and natural evolution of those for other values of x and ω, in contrast to the DEMA that, as noted above, fails dramatically for these values of the parameters.

The results of our investigation make we feel allowed to insert on the Bruggeman-Mie model our study of the effect of the interparticle correlation on the behaviour of the percolation of the composite referred to above. More precisely we intend to use the Bruggeman-Mie model with the specific purpose of

a) assessing how the introduction of a correlation of exclusion among the metal particles affects the properties of a composite;

b) studying the effect of the anisotropy of the particles.

To achieve our first goal, we forced onto the medium a correlation of exclusion that prevents the metal particles from touching each other by putting around each metal sphere a thin dielectric layer with $\varepsilon = 1$. The presence of the exclusion layer changes the topology of the dispersion from one typical of the aggregate structure model to another more akin to the separated grain model. Although our correlated model does not coincide completely with the classical Maxwell-Garnett model which is based on coated spheres only [1], while our model is based on a mixing of coated spheres and of dielectric spheres, our results show a gradual though still rather sharp decrease of the extinction when the thickness of the layer is increased.

In order to gain some information on the influence of the anisotropy of the scatterers on the optical behaviour of the composite, we force all the metal particles to aggregate in pairs thus producing effectively anisotropic scatterers. Of course, the aggregation itself could be considered as a kind of correlation forced onto the system. From this point of view, assuming that all the metal particles aggregate, the actual amount of correlation depends on either we consider the aggregates all oriented alike (*orientational correlation*) or at random. Our results for the metal-dielectric composite referred to above show that the aggregation shifts the percolation towards lower concentrations of the metallic component when the aggregates are randomly oriented; when the aggregates are all oriented alike larger shifts occur with their magnitude depending on the direction of incidence and on the polarization of the propagation wave. In practice it will be shown by discussing our actual results that the same orientation-dependent phenomena we found for single aggregates reappear in the composite at a collective level.

2. – Theory.

According to the effective medium theory ε_{eff}, the dielectric constant of the effective medium should satisfy the *null scattering condition*, *i.e.* it should be the solution to the equation [2]

$$\sum_i N_i f_{\eta,i}(x_i, \varepsilon_{\text{eff}}) = 0 \tag{1}$$

with

$$f_{\eta,i} = \hat{\boldsymbol{e}}_\eta^* \cdot \boldsymbol{f}_{\eta,i},$$

where $\hat{\boldsymbol{e}}_\eta$ is the polarization vector of the plane wave and $\boldsymbol{f}_{\eta,i}$ is the normalized forward-scattering amplitude [6, 7] of any of the N_i identical particles of the i-th kind, with size parameter x_i when immersed into the homogeneous effective medium of dielectric constant ε_{eff}.

It is perhaps worth noticing that the notation we will use throughout this paper differs from the customary one [8, 9]: it will conform, instead, to the one that was used in a paper the present authors [10] for it presents several advantages when dealing with aggregates of spheres.

Let us now assume that all the particles of the composite are spherical so that their forward-scattering amplitude is given by the Mie theory. Accordingly, for a homogeneous spherical particle with refractive index $n = \varepsilon^{1/2}$ embedded into a homogeneous medium of refractive index n_{ext}—in the present case $n_{\text{ext}} \equiv n_{\text{eff}} = \varepsilon_{\text{eff}}^{1/2}$—

$$f = -\frac{i}{4\pi k n_{\text{ext}}} \sum_{plm} W_{lm}^{(p)*} A_{lm}^{(p)}, \tag{2}$$

where the polarization index η has been dropped for the sake of simplicity, $k = \omega/c$ and

$$A_{lm}^{(p)} = -R_l^{(p)} W_{lm}^{(p)}. \tag{3}$$

In eqs. (2) and (3) the W's are the multipole amplitudes of the incident plane wave [7], the A's are those of the scattered field,

$$R_l^{(p)} = \frac{(1+\bar{n}\delta_{p1})\, U_l'\, u_l - (1+\bar{n}\delta_{p2}) U_l\, u_l'}{(1+\bar{n}\delta_{p1})\, U_l'\, v_l - (1+\bar{n}\delta_{p2}) U_l\, v_l'} \tag{4}$$

with

$$u_l = y_{\text{ext}} j_l(y_{\text{ext}}), \qquad v_l = y_{\text{ext}} h_l^{(1)}(y_{\text{ext}}), \qquad U_l = y j_l(y),$$

$$\bar{n} = \frac{n}{n_{\text{ext}}} - 1, \qquad y_{\text{ext}} = n_{\text{ext}} x, \qquad y = nx,$$

and $x = k\rho$ is the size parameter of the particle with radius ρ; p is a parity index that distinguishes the magnetic multipoles ($p = 1$) from the electric ones ($p = 2$), so that $R_l^{(1)}$ refers to the magnetic and $R_l^{(2)}$ to the electric 2^l-poles [7]. When $|y_{\text{ext}}| \ll 1$, eq. (4)

can be expanded and for $l=1$ one gets [9]

$$R_1^{(2)} \simeq -\frac{2}{3} i y_{\text{ext}}^3 \frac{\varepsilon - \varepsilon_{\text{ext}}}{\varepsilon + 2\varepsilon_{\text{ext}}}, \qquad R_1^{(1)} \simeq -\frac{i}{45} y_{\text{ext}}^5 \left(\frac{\varepsilon}{\varepsilon_{\text{ext}}} - 1 \right). \tag{5}$$

Since the $R_l^{(p)}$'s for $l>1$ are proportional to higher powers of y_{ext}, when dealing with small particles they are usually neglected. As a consequence, if one lets the size of the particles involved in eq. (1) tend to zero, only the $R_1^{(2)}$'s as given in eq. (5) need be considered and eq. (1) turns into the well-known Bruggeman mixing rule [1]

$$\sum_i V_i \frac{\varepsilon_i - \varepsilon_{\text{eff}}}{\varepsilon_i + 2\varepsilon_{\text{eff}}} = 0, \tag{6}$$

where $V_i = N_i v_i / V$ is the fraction of the total volume of the composite that is occupied by the N_i spheres of volume v_i and dielectric constant ε_i. Equation (6) is clearly independent of x_i and is thus a zero-size approximation to eq. (1). The dynamical effective medium approximation (DEMA) proposed by Stroud and Pan [2] is then easily obtained by including in the forward-scattering amplitudes also the $R_1^{(1)}$ term and thus taking into account also the magnetic-dipole effects. Since Stroud and Pan deal with small particles and thus take $R_1^{(1)}$ and $R_1^{(2)}$ still in the form of eq. (5), their mixing rule reads

$$\sum_i V_i \left[\frac{\varepsilon_i - \varepsilon_{\text{eff}}}{\varepsilon_i + 2\varepsilon_{\text{eff}}} + \frac{x_i^2}{30} (\varepsilon_i - \varepsilon_{\text{eff}}) \right] = 0. \tag{7}$$

Equation (7) is a frequency- and size-dependent extension of eq. (6) and can be applied to rather larger, although still small, particles. The mixing rule of Chýlek and Srivastava [5] lies on the very same approximations but, since the main concern of these authors is the effect due to a polidispersion of the metal particles, they take into account the size-dependent terms discussed above only when ε is very large—for instance, this is the case of metal spheres—but include a size-distribution function $\lambda(\rho)$ and write

$$\sum_i V_i \frac{\varepsilon_i - \varepsilon_{\text{eff}}}{\varepsilon_i + 2\varepsilon_{\text{eff}}} + \sum_{j;\, |\varepsilon_j| \gg 1} \frac{2\pi}{45} k^2 (\varepsilon_j - \varepsilon_{\text{eff}}) \int \rho^5 \lambda_j(\rho)\, \mathrm{d}\rho = 0. \tag{8}$$

Obviously, the size-dependent terms in eq. (8) assume the same form they have in eq. (7) when all the particles of the j-th kind have the same size, *i.e.* when $\lambda_j(\rho) = N_j \delta(\rho_j - \rho)$.

Since the only common implementations of eq. (1) are the zero-size approximation, eq. (6), and the small-size approximation, eqs. (7) and (8), in view of our purposes some testing of the reliability of the null scattering condition, eq. (1), when applied to larger particles is in order. To this end, all the calculations we will present in the next section were performed by carrying to full convergency the exact Mie expansions to the effect of ensuring the inclusion of all significant contributions to the scattering amplitude, eq. (2). In principle, our choice overcomes all the limitations due to the use of any small-size approximation, yet it is to be borne in mind that a dispersion of spheres with too big a size parameter can never be considered for the medium would then be so coarsely granulated as to prevent the possibility of treating the whole composite through a dielectric description.

2·1. *The correlation of exclusion.* – As mentioned in sect. 1, one of our main goals is the assessment of the effect of the interparticle correlation. Therefore, we considered a model composite medium which, in a sense, is a hybrid between that of Bruggeman and that of Maxwell-Garnett; namely we calculated ε_{eff} for a composite made of homogeneous dielectric spheres with $\varepsilon = \varepsilon_{\text{D}}$ and of spheres with a metal core, of dielectric constant ε_{M}, coated by a thin dielectric layer, still with $\varepsilon = \varepsilon_{\text{D}}$. The scattering properties of the coated spheres were not described through the approach based on the boundary conditions at the separation surface, but rather by interposing between the core and the coating a very thin transition layer [10] through which

$$\varepsilon = \varepsilon(r) = \varepsilon_{\text{M}} + (3s^2 - 2s^3)\Delta$$

with

$$\Delta = \varepsilon_{\text{D}} - \varepsilon_{\text{M}} \quad \text{and} \quad s = \frac{r - \rho_{\text{M}}}{\rho_{\text{D}} - \rho_{\text{M}}},$$

where ρ_{M} is the radius of the metal core and ρ_{D} the inner radius of the dielectric coating. In this way we achieved a smooth transition between the two materials as the refractive index $n = n(r)$ is a regular function of r everywhere within the coated spheres. Consequently eq. (4) must be substituted by [8]

$$R_l^{(p)} = \frac{G_l^{(p)\prime}\, u_l - (1 + \bar{n}\delta_{p2})^2 G_l^{(p)} u_l'}{G_l^{(p)\prime}\, v_l - (1 + \bar{n}\delta_{p2})^2 G_l^{(p)} v_l'}, \tag{9}$$

where $G_l^{(p)} = G_l^{(p)}(x)$ are now calculated by numerical integration of the radial equations

$$\frac{\mathrm{d}^2 G_l^{(p)}}{\mathrm{d}\xi^2} - \frac{2}{n}\frac{\mathrm{d}n}{\mathrm{d}\xi}\frac{\mathrm{d}G_l^{(2)}}{\mathrm{d}\xi}\delta_{p2} + \left[n^2 - \frac{l(l+1)}{\xi^2}\right] G_l^{(p)} = 0$$

with $\xi = kr$. It is worth noticing that, when the thickness of the coating reduces to zero, $\varepsilon(r) = \varepsilon_{\text{M}}$ and eq. (9) reduces to eq. (4), and also that our previous calculations [10] proved that the introduction of the transition layer, provided it is thin enough, has no influence on the results. As a matter of fact, the calculations on this model were performed by using eq. (9) with no approximation.

2·2. *The binary aggregation.* – In sect. 1 we mentioned the special kind of correlation that sets on into the space distribution when the particles undergo aggregation phenomena. Here we will briefly describe how these cases can be treated. Indeed, we presented elsewhere [11] an approach to the scattering properties of an aggregation of spheres (cluster) designed to yield the scattering properties of an aggregate as a whole. In particular the forward scattering amplitude of an aggregate is still in the form of eq. (2) but for that

$$A_{lm}^{(p)} = -\sum_{p'}\sum_{l'm'} S_{lm,l'm'}^{(p,p')} W_{l'm'}^{(p')}, \tag{10}$$

where the quantities $S_{lm,l'm'}^{(p,p')}$, defined in ref. [10], are analogous to the $R_l^{(p)}$s, eq. (3), for the single spheres and take account of the specific geometry of the aggregation as well as of its orientation. If, as often occurs, we have to deal with a dispersion of ran-

domly oriented identical aggregates and $f(\Theta)$ is the scattering amplitude of any one of them with orientation Θ, we can sum over all the orientations and get

$$\frac{N}{8\pi^2}\int f(\Theta)\,\mathrm{d}\Theta = N\bar{f},$$

where $\bar{f}$ on the right-hand side is still in the form of eq. (2) but in this case the amplitudes A should be substituted by their orientational averages $\bar{A}$ [12] given by

$$\bar{A}_{lm}^{(p)} = -\frac{1}{2l+1}\sum_{p'}\left[\sum_{m''} S_{lm''lm''}^{(p,p')}(\Theta_0)\right]W_{lm}^{(p')}. \tag{11}$$

Thanks to the structure of the summation in eq. (11), the resulting $\bar{A}$'s are independent of the particular orientation, Θ_0, we chose to perform the calculation. The quantities S in eqs. (10) and (11) are the elements of a matrix that must be obtained from the elements of the inverse of a larger matrix, defined in ref. [12], of order $2\mathcal{N}L(L+2)$, where $\mathcal{N}$ is the number of the spheres in the aggregate and L is the value of l at which the sums are truncated. Considerations on the symmetry of the aggregate, if any, can be used to factorize the matrix to be inverted [13, 14]; in this respect, the most convenient case occurs when the centres of the spheres lie on a straight line, for the maximum order of the matrices to be inverted is then $2\mathcal{N}L$ only.

3. – Results and discussion.

Let us recall that the composite we dealt with in our calculations is a two-component one; the first component is a dielectric with $\varepsilon_D = 1$ and the second is a met-

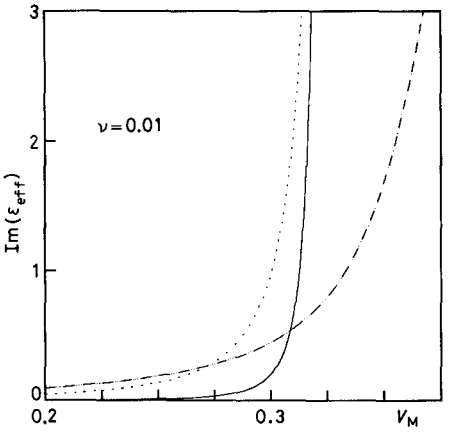

Fig. 1. – $\mathrm{Im}[\varepsilon_{eff}]$ *vs.* V_M, for $\nu = 0.01$ and $x_D = 10^{-4}$. The dotted curve reports the results of the Bruggeman-Mie model for $x_M = 0.01$, while the dashed curve reports those for $x_M = 0.1$. On this scale the results from the DEMA for $x_M = 0.01$ coincide with the dotted curve. We also notice that the DEMA does not give acceptable results for $x_M = 0.1$ (see the text). The solid curve reports for comparison the predictions of the classical Bruggeman rule.

al with

$$\varepsilon_M = 1 - \frac{1}{\nu(\nu + i\gamma)}, \tag{12}$$

where $\nu = \omega/\omega_p$, $\gamma = 1/\tau\omega_p$, *viz.* the dielectric material is the vacuum while the metallic component is described by the free-electron Drude constant [15]. As a matter of fact, in eq. (12) we always choose $\gamma = 0.01$.

If we assume the particles to be spherical, eq. (1) becomes

$$V_M f_M(\varepsilon_{eff}) + (\rho_M/\rho_D)^3 (1 - V_M) f_D(\varepsilon_{eff}) = 0, \tag{13}$$

where ρ_D is the radius of the metal spheres and ρ_D the radius of the dielectric ones, and the f's are given by eqs. (2), (3) and (4). It is perhaps worth noticing that eq. (13), unlike the two-component implementations of eqs. (6)-(8), is a transcendental equation in ε_{eff}, and thus requires some caution in seeking for its solutions. Anyway, we calculated ε_{eff} according to eq. (13) as a function of V_M for some values of $x_M = k\rho_M$ and of ν. In particular in fig. 1 and 2 we report $\mathrm{Im}[\varepsilon_{eff}]$ as calculated from eq. (13) and, for the sake of comparison, from the two-component implementations of eq. (6) and of eq. (7). We chose to plot our results for $\nu = 0.01$, fig. 1, and for $\nu = 0.1$, fig. 2, since for most metals these frequencies occur in the far and in the near infrared, respectively. For instance, for aluminium ($\omega_p = 2.4 \cdot 10^{16}\,\mathrm{s}^{-1}$) [16] these values of ν correspond to $\lambda \approx 8\,\mu\mathrm{m}$ and $0.8\,\mu\mathrm{m}$, respectively.

In fig. 1 the dotted and the dashed curves report the results of the Bruggeman-Mie model for $x_M = 0.01$ and 0.1, respectively. The results of the classical Bruggeman mixing rule are given by the solid curve for the sake of comparison. For $x_M = 0.01$ the predictions of the DEMA coincide, on the scale of the figure, with the dotted curve thus proving that the DEMA is adequate for this size. The situation is quite different for $x_M = 0.1$: for this size, indeed, the DEMA gives results that are physically unacceptable and are therefore not reported in fig. 1. In fact the DEMA,

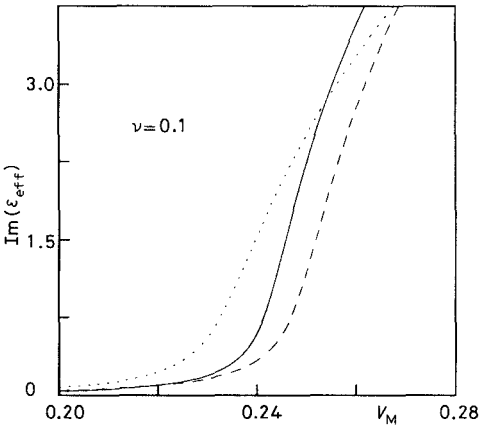

Fig. 2. – $\mathrm{Im}[\varepsilon_{eff}]$ *vs.* V_M, for $\nu = 0.1$ and $x_D = 10^{-4}$. The solid and the dotted curves report the results of the Bruggeman-Mie model for $x_M = 0.01$ and $x_M = 0.1$, respectively. On the scale of the figure the predictions of the DEMA for $x_M = 0.01$ and of the classical Bruggeman rule are indistinguishable from each other and from the solid curve. The results from the DEMA for $x_M = 0.1$ are reported by the dashed curve. Note that the Bruggeman-Mie model for $x_M = 0.1$ predicts the lowest value of V_M for the onset of the percolation phenomena.

that is an algebraic equation of the third degree, yields values of $\mathrm{Im}[\varepsilon_{\mathrm{eff}}]$ that are either negative or everywhere vanishingly small or very large even in the low-V_{M} region. This behaviour, on account of the acceptability of the corresponding solutions from the Bruggeman-Mie model, is easily attributable to the small-size approximations on which the DEMA itself is based.

In fig. 2 the solid curve reports the predictions of the Bruggeman-Mie model for $x_{\mathrm{M}} = 0.01$ which, on the scale of the figure, coincide with the predictions of the DEMA for the same value of x_{M} and with those of the classical Bruggeman model. The dotted curve, in turn, reports the results of the Bruggeman-Mie model for $x_{\mathrm{M}} = 0.1$ while the dashed curve reports those of the DEMA for the same value of x_{M}. For $x_{\mathrm{M}} = 0.1$ our results predict the onset of the percolation phenomena at the lowest value of V_{M}. In our opinion this is a clear indication that at $x_{\mathrm{M}} = 0.1$ there is a lack of convergence in both the DEMA and in the zero size approximation. On the other hand, the value $\varepsilon_{\mathrm{eff}}$ assumes at $\nu = 0.1$ makes the results from the DEMA to appear in the figure, in contrast to what happens for $\nu = 0.01$ (see fig. 1).

We feel it necessary, at this stage, to spend some words to point out that both our equation (13) and its counterpart according to Stroud and Pan, eq. (7), are size dependent; in particular they depend on the size of the dielectric particles. When one assumes $\varepsilon_{\mathrm{D}} = 1$, *i.e.* when the dielectric is the vacuum and not an actual medium, the problem arises whether a more or less minute subdivision of the former into spherical cells may affect the results. As a matter of fact, the curves in fig. 1 and 2 were obtained by taking $x_{\mathrm{D}} = k\rho_{\mathrm{D}} = 10^{-4}$ but careful test calculations allow us to state that the size of the vacuum spheres has very small influence either when it is set to zero according to Chýlek and Srivastava or when it is set equal to the size of the metal spheres as in the paper of Stroud and Pan. In other words, for the values of x_{M} and V_{M} we considered in our calculations, the Bruggeman-Mie model yields results that are independent of x_{D} while the results from the DEMA for $x_{\mathrm{D}} \approx x_{\mathrm{M}}$ are undistinguishable from those of Chýlek and Srivastava with a δ-like distribution of the radii. The independence of x_{D} was to be expected because, when $|\varepsilon_{\mathrm{M}}| \gg 1$, the contribution to the scattering amplitude from the size-dependent terms of the dielectric particles becomes fairly small in comparison to the corresponding contributions of the metal spheres. This is true as far as $\varepsilon_{\mathrm{eff}}$ does not become in turn so large as to enhance the size-dependent contributions from the dielectric spheres: this is easily understood *e.g.* from the size-dependent term of eq. (7). Anyway, the range of V_{M} in all the figures was chosen so as to show the influence of the size and of the frequency on the behaviour of $\varepsilon_{\mathrm{eff}}$ in the region in which it becomes rather large, for just in this region the composite undergoes the transition from the dielectric behaviour to the metallic one. On the basis of the results depicted in fig. 1 and 2, $x_{\mathrm{M}} = 0.1$ seems to be an upper limit for the usefulness of eq. (13) for our purposes as it excludes from the results any unwanted x_{D}-dependence. Nevertheless we found that, when the Bruggeman-Mie model is applied to particles of size greater that 0.1 but rather smaller than 1, the form of $\mathrm{Im}[\varepsilon_{\mathrm{eff}}]$ from eq. (13) is still meaningful in the small-V_{M} region, *i.e.* before the onset of the percolation phenomena.

3·1. *The correlation of exclusion.* – When the metal spheres are coated by a dielectric layer we have

$$V_{\mathrm{M}} f_{\mathrm{MC}}(\varepsilon_{\mathrm{eff}}) + (\rho_{\mathrm{M}}/\rho_{\mathrm{D}})^3 [1 - V_{\mathrm{M}}(1 + c_{\mathrm{L}})] f_{\mathrm{D}}(\varepsilon_{\mathrm{eff}}) = 0\,, \tag{14}$$

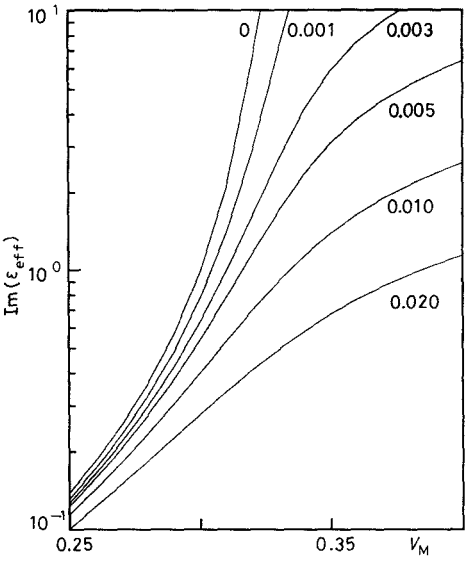

Fig. 3. – $\mathrm{Im}[\varepsilon_{\mathrm{eff}}]$ *vs.* V_{M}, from eq. (14), calculated for the values of c_{L} that label the curves. $x_{\mathrm{M}} = 0.01$, $x_{\mathrm{D}} = 10^{-4}$ and $\nu = 0.01$.

where f_{MC} is the forward-scattering amplitude of a sphere with a metal core of radius ρ_{M} and a dielectric coating layer with external radius

$$\rho_{\mathrm{L}} = (1 + c_{\mathrm{L}})^{1/3} \rho_{\mathrm{M}} ,$$

and is thus given by eqs. (2), (3) and (9). The quantity c_{L} is defined as

$$c_{\mathrm{L}} = v_{\mathrm{L}} / v_{\mathrm{M}} ,$$

where v_{L} is the volume of the dielectric layer: in other words c_{L} gives the volume of the dielectric coating around each metal sphere as a fraction of the volume, v_{M}, of the metal spheres themselves. Of course

$$V_{\mathrm{M}} \leqslant V_{\mathrm{M\,max}} = \frac{1}{1 + c_{\mathrm{L}}} , \qquad 0 \leqslant c_{\mathrm{L}} \leqslant c_{\mathrm{L\,max}} = \frac{1}{V_{\mathrm{M}}} - 1 ,$$

for a given c_{L} and for a given V_{M}, respectively, so that, when $c_{\mathrm{L}} = 0$, *i.e.* when the spheres have no coating, we get again the topology of the Bruggeman model, whereas, when $V_{\mathrm{M}} = V_{\mathrm{M\,max}}$ or $c_{\mathrm{L}} = c_{\mathrm{L\,max}}$, we get the Maxwell-Garnett topology.

In fig. 3 and 4 we plot $\mathrm{Im}[\varepsilon_{\mathrm{eff}}]$ for several values of c_{L} as calculated according to eq. (14) for $x_{\mathrm{M}} = 0.01$ and 0.1, respectively. Only the results for $\nu = 0.01$ are shown because those for $\nu = 0.1$ are strictly analogous. It is quite evident that the correlation of exclusion produced by the dielectric coating around the metal spheres has very strong effects even at very low values of c_{L}. Actually, the model described in eq. (14) prevents any contact among the metal particles and is therefore topologically akin to the Maxwell-Garnett model. As a consequence, it is not surprising that, when c_{L} increases, our results tend very fast to those that are typical of the latter model. Figures 3 and 4 show the results for $x_{\mathrm{D}} = 10^{-4}$. At this value of x_{D} the contribution of the vacuum spheres to the scattering amplitude is very well approximated by the first of

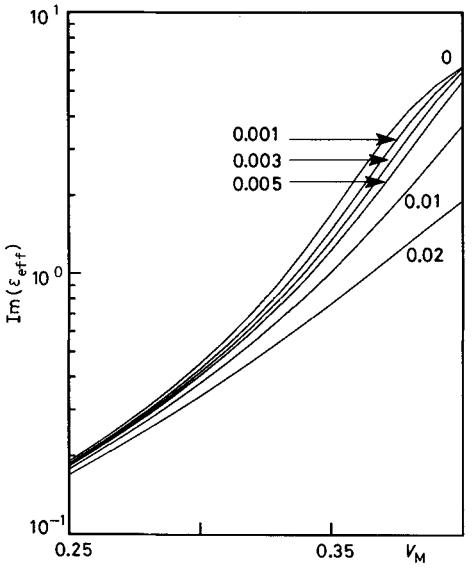

Fig. 4. – Same as fig. 3 but for $x_M = 0.1$.

equations (5) and such a small size suggests the idea of a dielectric medium which pervades and fills all the space around the coated spheres. Even in this case the choice of the size of the dielectric spheres, up to $x_D = x_M$, does not affect the results of our calculations from eq. (14). The different rate of diminution of the extinction coefficient in fig. 3 and 4 should be noticed. The lower rate of decrease in fig. 4 is in fact due to the lower value of the ratio $|\varepsilon_M/\varepsilon_D|$ for the higher values of ν. On the whole, we feel that the correlation forced into the medium by coating all the metal particles is a bit excessive: a more flexible model than that of eq. (14) can, indeed, be built by introducing a distribution function for the thickness of the dielectric coating, including the possibility of zero thickness. Such a model would prevent the contact among (only a fraction of) the metal spheres, so that an appropriate choice of the distribution function should ensure a smoother transition from the results of the Bruggeman model and those of the Maxwell-Garnett model. In this respect let us point out that the electron microscopy reveals that in some dispersions of metal particles several of them appear as coated by their oxide [17].

3·2. *Effect of the aggregation.* – When the metal spheres aggregate in pairs, we have

$$V_M f_C(\varepsilon_{\mathrm{eff}}) + 2(\rho_M/\rho_D)^3 [1 - V_M] f_D(\varepsilon_{\mathrm{eff}}) = 0\,, \tag{15}$$

where f_C is the contribution of one aggregate, possibly averaged over the orientations as described in the preceding section.

Figures 5 and 6, where $\mathrm{Im}[\varepsilon_{\mathrm{eff}}]$ is plotted from eq. (15) for $\nu = 0.01$, show the effect of a complete binary aggregation of the metal particles on the transition of a composite medium to the metallic behaviour; analogous results were obtained for $\nu = 0.1$. Even in this case the value of x_D proves to be irrelevant, although the conceptually most satisfactory choice is that of very small size (in practice $x_D = 10^{-4}$). The effect of

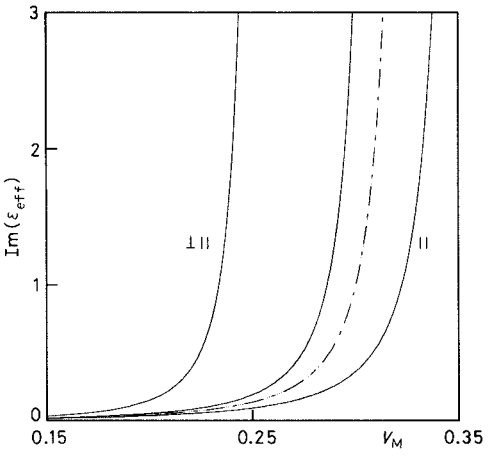

Fig. 5. – Im[ε_{eff}] *vs.* V_M, from eq. (15), for binary aggregates with $x_M = 0.01$, $x_D = 10^{-4}$ and $\nu = 0.01$. The unlabelled solid curve refers to randomly oriented aggregates; that labelled ‖ refers to aggregates all oriented with their axes parallel to the wavevector and the one labelled ⊥ ‖ refers to aggregates with their axes parallel to the electric field of the wave which is assumed to be linearly polarized; notice the low value of the percolation threshold predicted by this latter curve. The dashed line reports, for the sake of comparison, the results from eq. (13).

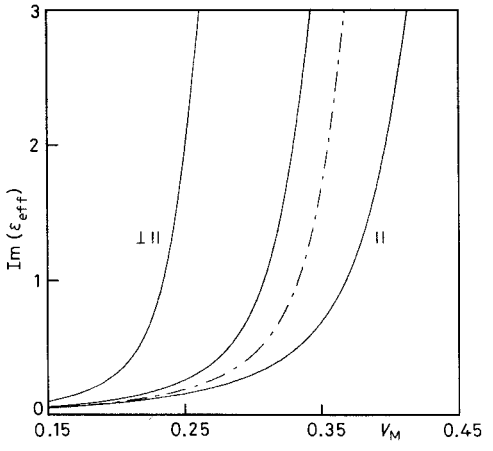

Fig. 6. – Same as fig. 5 but for $x_M = 0.1$.

the aggregation proves to be a quite significant one but it is very strongly increased when to the correlation of aggregation one also adds the request that all the binary aggregates be likely oriented. Both figures 5 and 6 show that the percolation undergoes a noticeable shift towards higher values of V_M when the common orientation of the aggregates is chosen so as their axes are parallel to the wavevector of the incident wave. This phenomenon is easily understandable if one thinks of the arrangement above as the one that provides the maximum of metal-free paths to the propagation radiation. On the other hand, the maximum in the extinction power of the medium that occurs when the axes of the aggregates are all parallel to the electric field can also be attributed to the arrangement, as this choice of the orientation maximizes the

absorption of the individual aggregates. These latter results prove that the anisotropy of the metal particles plays a fundamental role in determining the optical properties of the composite. It may be important to notice that the onset of the percolation effects is predicted in this case at $V_M \approx 0.22$ both for $x_M = 0.01$ and $x_M = 0.1$. This suggests that in the determination of the experimental data the aggregation of the metallic domains plays a relevant role. In this respect let us recall that Webman, Jortner and Cohen state that the experimental evidence supports the occurrence of clustering in alkali-tungsten bronzes and in several other composites [4]. Less strong effects than those shown in fig. 5 and 6 should be expected if, more realistically, only a reasonable fraction of the metal particles were forced to aggregate.

3·3. *Conclusions.* – The results we discussed above show that our Bruggeman-Mie approach yields meaningful results even when applied to particles with $x \simeq 0.1$ and in this respect proves superior to the customary approaches, eqs. (6)-(8). Unfortunately neither the Bruggeman-Mie model is able to predict the onset of the percolation effects in agreement with the experimental data. Bearing in mind the conditions stated by Webman, Jortner and Cohen and by Stroud and Pan for the validity of the effective medium approximation this is in no way surprising, because these conditions can hardly by satisfied by a metal-dielectric composite but at frequencies near ω_p. Anyway, the stability of the results when all the nonessential approximations (such as the small-size one) are removed strongly suggests that our investigation of the effects of the correlation and of the aggregation can be legitimately superposed on the Bruggeman-Mie model. Of course, as anticipated in sect. **1**, the shifts of the percolation we attribute to the correlation or to the aggregation must be understood in a relative sense. In turn, our attempt to introduce some correlation among the metal particles, eqs. (14) and (15), proved successful. As remarked above, however, since there are no experimental data on the exact amount of correlation to be forced onto specific composites we chose to assume that *all* the particles were strongly correlated both in the case of exclusion and in that of aggregation. As a result the amount of correlation we introduced into our model turns out to be so large that the results we reported can give only semi-quantitative information, *i.e.* they can indicate the trend one has to expect in the more realistic situations that could be effectively simulated by assuming *e.g.* that only a fraction of the metal particles are coated or by forcing only a fraction of the particles to aggregate. Anyway, the models we presented in this paper can give, in our opinion, useful indications on the dielectric behaviour of the composite media.

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